

SAT Hard Question 16

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Question: Consecutive Integers and Difference Condition

For a set of three consecutive integers, the result of subtracting the greatest integer from the sum of the other two integers is 48. What is the value of the greatest of the three integers?

Answer Explanation: Algebraic Setup and Solution

Let the three consecutive integers be n , $n + 1$, and $n + 2$.
Sum of the smaller two integers:

$$n + (n + 1) = 2n + 1$$

Subtract the greatest integer:

$$2n + 1 - (n + 2) = 2n + 1 - n - 2 = n - 1$$

Set equal to 48:

$$n - 1 = 48$$

$$n = 49$$

The greatest integer is:

$$n + 2 = 49 + 2 = \boxed{51}$$

Question: Quadratic Function and Ground Impact Time

The function $h(t) = -16t^2 + b$ estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant.

The function estimates that the object is 40.96 feet above the ground when it is dropped at $t = 0$.

How many seconds after being dropped does the function estimate the object will hit the ground?

Answer Explanation: Solving the Quadratic Equation

Since $t = 0$ gives $h(0) = b = 40.96$, the function is:

$$h(t) = -16t^2 + 40.96$$

Set $h(t) = 0$ to find when the object hits the ground:

$$-16t^2 + 40.96 = 0$$

$$16t^2 = 40.96$$

$$t^2 = \frac{40.96}{16} = 2.56$$

$$t = \sqrt{2.56} = 1.6$$

The object hits the ground after:

$$\boxed{1.6} \text{ seconds}$$

Question: Survey Percentages and Totals

A researcher surveyed undergraduate students, graduate students, and postdoctoral students. The number of undergraduate students surveyed was 7,150% of the number of postdoctoral students surveyed, and the number of graduate students surveyed was 45% of the number of undergraduate students surveyed.

If there were 6,435 graduate students surveyed, what was the sum of the number of undergraduate students and postdoctoral students surveyed?

Answer Explanation: Setting Up Proportions and Solving

Let the number of postdoctoral students be x .

Number of undergraduate students:

$$71.5x$$

Number of graduate students:

$$0.45 \times 71.5x = 32.175x$$

Set equal to 6,435:

$$\begin{aligned} 32.175x &= 6435 \\ x &= \frac{6435}{32.175} = 200 \end{aligned}$$

So, number of undergraduate students:

$$71.5 \times 200 = 14,300$$

Sum of undergraduate and postdoctoral students:

$$14,300 + 200 = \boxed{14,500}$$

Question: Parallel Lines Condition for No Solution

$$16x - 20y = 6y + 8$$

$$ty = 8x + \frac{1}{5}$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t ?

Answer Explanation: Comparing Slopes for No Solution Condition

Rewrite both equations in slope-intercept form.

First equation:

$$16x - 20y = 6y + 8$$

$$16x - 20y - 6y = 8$$

$$16x - 26y = 8$$

$$y = \frac{16}{26}x - \frac{8}{26} = \frac{8}{13}x - \frac{4}{13}$$

Second equation:

$$ty = 8x + \frac{1}{5}$$

$$y = \frac{8}{t}x + \frac{1}{5t}$$

For no solution, slopes must be equal:

$$\frac{8}{13} = \frac{8}{t}$$

$$t = 13$$

Question: Unit Conversion of Bone Mineral Density

The bone mineral density of a certain person's thoracic spine was estimated to be 0.812 grams per square centimeter.

What is this estimated bone mineral density in grams per square millimeter? (1 centimeter = 10 millimeters)

Answer Explanation: Converting Area Units

$$1 \text{ cm}^2 = (10 \text{ mm})^2 = 100 \text{ mm}^2$$

So,

$$0.812 \text{ g/cm}^2 = \frac{0.812}{100} \text{ g/mm}^2 = 0.00812 \text{ g/mm}^2$$

Thus, the answer is:

0.00812

Question: Product of Roots and Factored Form

$$\frac{5}{9}(5x + 9)(x + \sqrt{5k + 9})(x - \sqrt{5k + 9}) = 0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 72. What is the value of k ?

Answer Explanation

For the solution

$$x = -\frac{9}{5}, -\sqrt{5k+9}, \text{ and } \sqrt{5k+9}$$

The product of the solution is

$$\frac{9}{5}(5k+9) = 72$$

Where $k = 6.2$

Question: Surface Area of a Right Rectangular Prism

A right rectangular prism has a base area of $56t$ square centimeters. The length of the base is $\frac{14}{3}$ cm, and the height is 3 cm.
What is the expression of the surface area, in cm^2 , of the prism?

Answer Explanation: Surface Area Calculation

$$184t + 28$$

Question: Unit Circle, Coordinates, and Trigonometric Ratio

In the xy -plane, a unit circle with center at the origin O contains point A at $(1, 0)$ and point B at $\left(\frac{5}{\sqrt{34}}, -\frac{3}{\sqrt{34}}\right)$.

If $\angle AOB$ measures p radians, what is the value of $\frac{\cos p}{\sin p}$?

Answer Explanation

We are asked to compute:

$$\frac{\cos p}{\sin p}$$

Recall the identity:

$$\frac{\cos p}{\sin p} = \cot p = \frac{1}{\tan p}$$

Point B has coordinates:

$$\left(\frac{5}{\sqrt{34}}, -\frac{3}{\sqrt{34}} \right)$$

So:

$$\tan p = \frac{\text{y-coordinate}}{\text{x-coordinate}} = \frac{-\frac{3}{\sqrt{34}}}{\frac{5}{\sqrt{34}}} = -\frac{3}{5}$$

Thus:

$$\frac{\cos p}{\sin p} = \frac{1}{\tan p} = \frac{1}{-\frac{3}{5}} = -\frac{5}{3}$$

The value is:

$$\boxed{-\frac{5}{3}}$$

Question: Maximum Root of a Quadratic Function

The function h is defined by:

$$h(x) = -\sqrt{x^2 + bx + c}$$

where b and c are constants. In the xy -plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{334})$.

If $h(m) = 0$, what is the greatest possible value of m ?

Answer Explanation: Solving for Maximum Root

Thus, the greatest possible value of m is:

167

Question: Linear Cost Model for Computer Repair

A computer repair specialist charges \$160 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 6 hours of repair is \$360. What is the function f that gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

Answer Explanation: Constructing the Piecewise Linear Model

The first two hours cost \$160. For 6 hours, the total cost is \$360, so the cost of the 4 additional hours is:

$$360 - 160 = 200$$

Hourly fee:

$$\frac{200}{4} = 50$$

Model:

$$f(x) = 160 + 50(x - 2) = 50x + 60$$

Thus, the function is:

$$f(x) = 50x + 60$$